Bike Rental Management System

Database Management System Project

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Project Overview

Purpose of the Bike Rental Management System:

Manage customer bike rentals efficiently

Store customer, booking, payment, and maintenance records

Track bike availability and rental status

Manage partner-owned bikes and commission system

Improve operational efficiency through centralized database

Problem Statement

Issues encountered without a dedicated database system:

Manual booking errors

Difficult bike tracking

Missing payment records

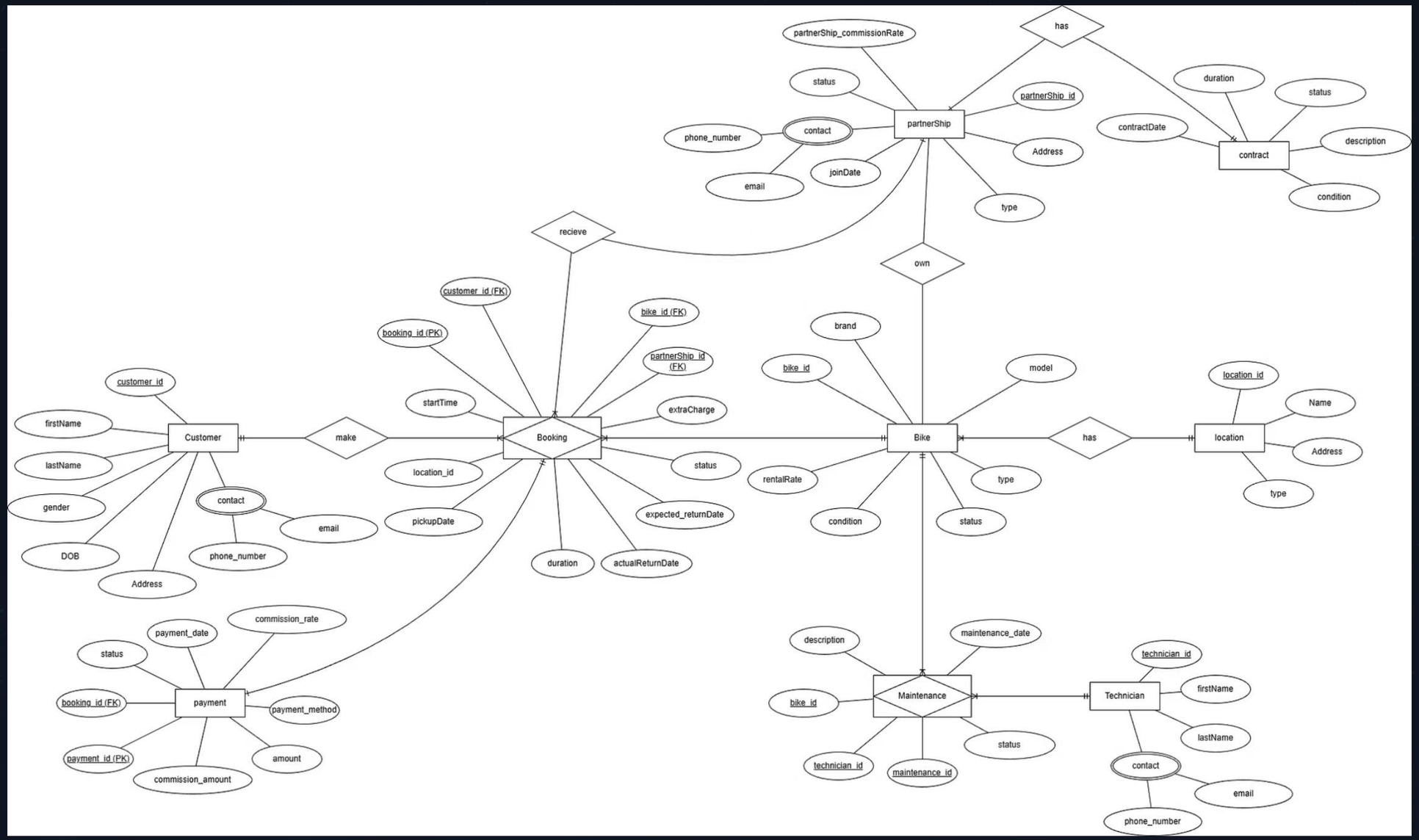
Poor maintenance scheduling

Inefficient customer management

Project Objectives

- Design normalized database structure
- Build Entity Relationship Diagram (ERD)
- Convert ERD into Relational Model
- Implement database using MySQL
- Support payment and commission tracking

Entity Relationship Diagram (ERD)



ERD Explanation - Main Entities

Customer

Booking

Bike

Partnership

Payment

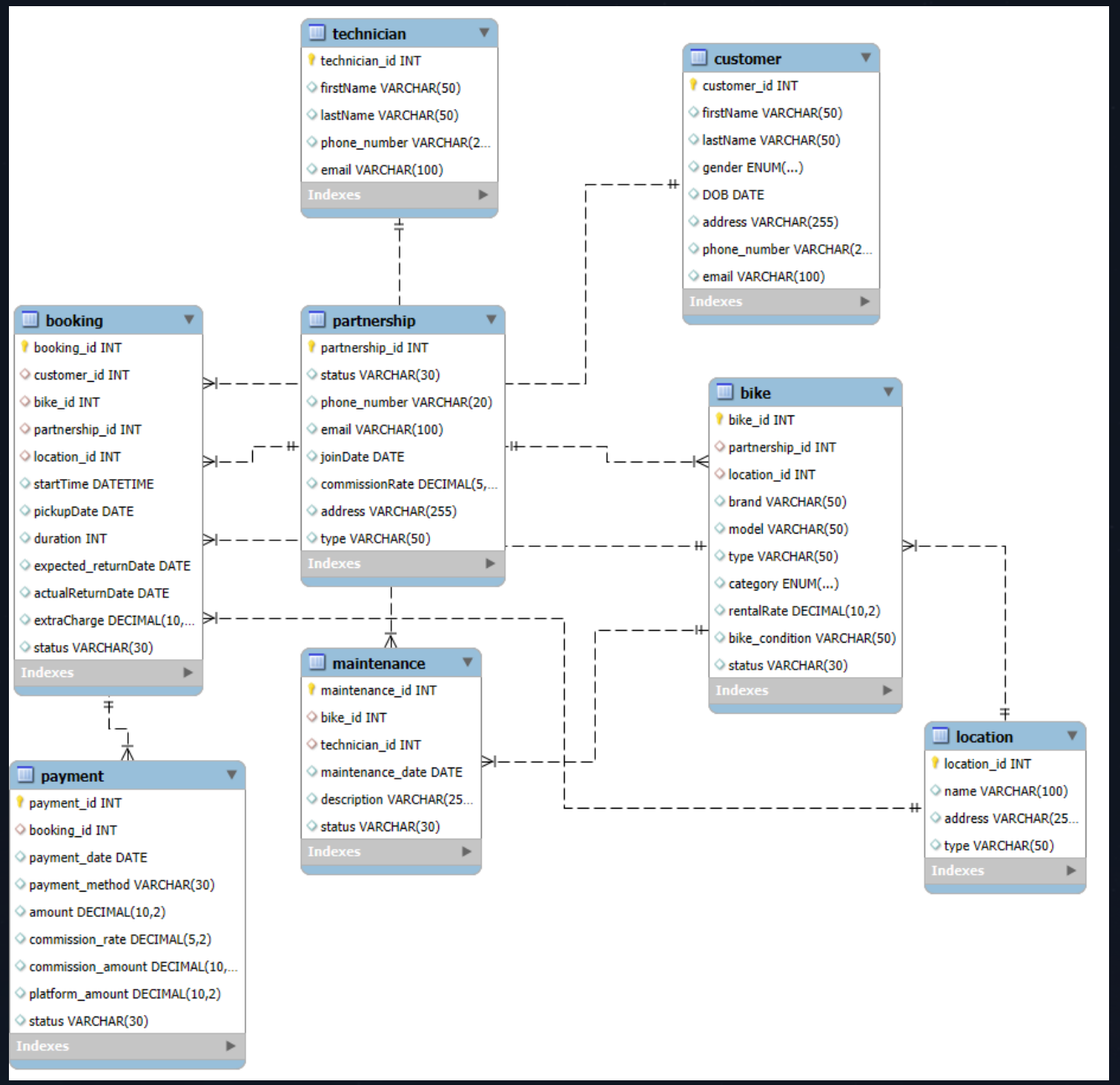
Maintenance

Technician

Location

Contract

Relational Model / Database Schema



Key Relationships

Customer → Booking

(1:M)

Bike → Booking

(1:M)

Partnership → Bike

(1:M)

Booking → Payment

(1:1)

Bike → Maintenance

(1:M)

Technician → Maintenance

(1:M)

SQL Implementation (DDL)

- CREATE DATABASE
- CREATE TABLE
- PRIMARY KEY
- FOREIGN KEY
- Constraints

Example DDL Statements:

```
CREATE DATABASE bike_rental_db;

USE bike_rental_db;

CREATE TABLE Customers (
    customer_id INT PRIMARY KEY AUTO_INCREMENT,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    phone_number VARCHAR(15),
    CONSTRAINT CHK_phone_number CHECK (LENGTH(phone_number) >= 10)
);

CREATE TABLE Bikes (
    bike_id INT PRIMARY KEY AUTO_INCREMENT,
    model VARCHAR(50) NOT NULL,
    type VARCHAR(50),
    rental_rate DECIMAL(10, 2) NOT NULL,
    status ENUM('available', 'rented', 'maintenance') DEFAULT 'available',
    partner_id INT,
    FOREIGN KEY (partner_id) REFERENCES Partnerships(partner_id)
);
```

SQL Operations (DML & DQL)

DML

```
INSERT INTO Customers (first_name, last_name,
email, phone_number)
VALUES ('John', 'Doe', 'john.doe@example.com',
'5551234567');

UPDATE Bikes
SET status = 'rented'
WHERE bike_id = 101;

DELETE FROM Bookings
WHERE booking_status = 'cancelled';
```

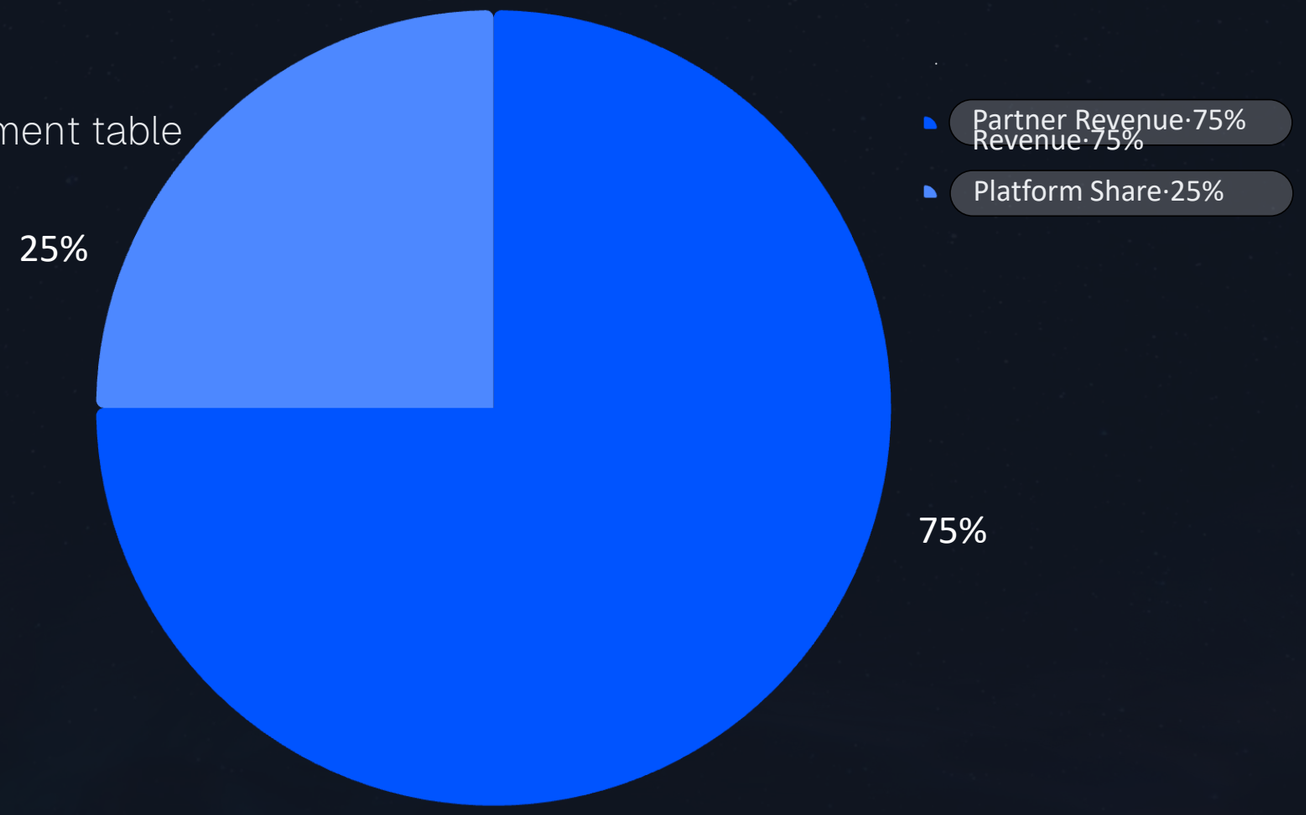
DQL (Data Query Language)

```
SELECT bike_id, model, type, rental_rate
FROM Bikes
WHERE status = 'available';

SELECT c.customer_id, c.first_name, c.last_name,
b.booking_id, b.start_date, b.end_date
FROM Customers c
JOIN Bookings b ON c.customer_id = b.customer_id
WHERE c.customer_id = 1001;

SELECT SUM(total_amount) - SUM(expenses) AS
total_company_profit
FROM Financials;
```

Commission System

Our commission system is designed to provide a fair and transparent revenue sharing model for our partners:

- Partner owns bike
- Partner receives 75% rental revenue
- Platform keeps 25%
- Automatically calculated in Payment table



Conclusion

Successfully designed complete database system

Supports rentals, payments, maintenance, partnerships

Improves data accuracy and efficiency

Scalable for future mobile/web application

Questions?

Thank You